

Real Analysis

A Self-Contained Set of Lecture Notes

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Abstract

These notes cover the core topics of a first graduate course in real analysis: measure theory, the Lebesgue integral, L^p spaces, and Fourier analysis. All results are stated precisely and most are proved in full.

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1 Measure Theory

1.1 σ -Algebras and Measures

Definition 1.1 $\triangleright$ σ -Algebra

Let X be a non-empty set. A collection $\mathcal{M} \subseteq \mathcal{P}(X)$ is called a σ -algebra on X if

1. $X \in \mathcal{M}$,
2. $A \in \mathcal{M} \Rightarrow A^c \in \mathcal{M}$,
3. $\{A_n\}_{n=1}^{\infty} \subseteq \mathcal{M} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{M}$.

The pair $(X, \mathcal{M})$ is called a *measurable space*.

Definition 1.2 $\triangleright$ Measure

A function $\mu : \mathcal{M} \rightarrow [0, +\infty]$ is a *measure* on $(X, \mathcal{M})$ if

1. $\mu(\emptyset) = 0$, and
2. (*countable additivity*) for every sequence of pairwise disjoint sets $\{A_n\} \subseteq \mathcal{M}$,

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n).$$

The triple $(X, \mathcal{M}, \mu)$ is a *measure space*.

Theorem 1.3 $\triangleright$ Continuity of Measure

Let $(X, \mathcal{M}, \mu)$ be a measure space.

1. (Upward continuity) If $A_1 \subseteq A_2 \subseteq \dots$ in $\mathcal{M}$, then $\mu(\bigcup_n A_n) = \lim_{n \rightarrow \infty} \mu(A_n)$.
2. (Downward continuity) If $A_1 \supseteq A_2 \supseteq \dots$ in $\mathcal{M}$ and $\mu(A_1) < \infty$, then $\mu(\bigcap_n A_n) = \lim_{n \rightarrow \infty} \mu(A_n)$.

Proof. (1) Define $B_1 = A_1$ and $B_n = A_n \setminus A_{n-1}$ for $n \geq 2$. The sets $\{B_n\}$ are pairwise disjoint and $\bigcup_{k=1}^n B_k = A_n$, so by countable additivity

$$\mu\left(\bigcup_{n=1}^{\infty} A_n\right) = \mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} \mu(B_n) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mu(B_k) = \lim_{n \rightarrow \infty} \mu(A_n).$$

Borel
 σ -algebra:
generated by
open sets of
a topology.

| (2) Apply part (1) to $C_n = A_1 \setminus A_n$, noting $\mu(A_1) < \infty$. □

2 The Lebesgue Integral

2.1 Construction

Definition 2.1 ▷ Measurable Function

Let $(X, \mathcal{M})$ and $(Y, \mathcal{N})$ be measurable spaces. A function $f : X \rightarrow Y$ is *measurable* if $f^{-1}(B) \in \mathcal{M}$ for every $B \in \mathcal{N}$.

Lemma 2.2 ▷ Monotone Convergence Theorem

Let $\{f_n\}$ be a sequence of non-negative measurable functions with $f_n \nearrow f$ pointwise. Then

$$\lim_{n \rightarrow \infty} \int f_n \, d\mu = \int f \, d\mu.$$

Theorem 2.3 ▷ Dominated Convergence Theorem

Let $\{f_n\}$ be measurable functions with $f_n \rightarrow f$ μ -a.e. If there exists $g \in L^1(\mu)$ such that $|f_n| \leq g$ μ -a.e. for all n , then $f \in L^1(\mu)$ and

$$\lim_{n \rightarrow \infty} \int f_n \, d\mu = \int f \, d\mu.$$

Proof. Since $g \pm f_n \geq 0$ and $g \pm f_n \rightarrow g \pm f$ μ -a.e., Fatou's lemma gives

$$\int (g \pm f) \, d\mu \leq \liminf_{n \rightarrow \infty} \int (g \pm f_n) \, d\mu.$$

Subtracting $\int g \, d\mu < \infty$ and combining the two inequalities yields $\limsup \int f_n \, d\mu \leq \int f \, d\mu \leq \liminf \int f_n \, d\mu$. □

Corollary 2.4

Under the hypotheses of ?? 2.3, $\|f_n - f\|_{L^1} \rightarrow 0$ as $n \rightarrow \infty$.

3 L^p Spaces

Definition 3.1 ▷ *L^p Space*

For $1 \leq p < \infty$ define

$$L^p(\mu) = \left\{ f \text{ measurable} : \|f\|_p := \left(\int |f|^p d\mu \right)^{1/p} < \infty \right\}.$$

For $p = \infty$, $\|f\|_\infty := \text{ess sup}|f|$.

Theorem 3.2 ▷ *Hölder's Inequality*

Let $p, q \in (1, \infty)$ be conjugate exponents, $\frac{1}{p} + \frac{1}{q} = 1$. For $f \in L^p$, $g \in L^q$,

$$\int |fg| d\mu \leq \|f\|_p \|g\|_q.$$

Example 3.3 ▷ *Convolution on $\mathbb{R}^n$*

For $f \in L^1(\mathbb{R}^n)$ and $g \in L^p(\mathbb{R}^n)$ the convolution

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x - y) g(y) dy$$

satisfies $\|f * g\|_p \leq \|f\|_1 \|g\|_p$ (Young's inequality).

Remark 3.4 ▷ *On completeness*

Fischer–Riesz: $L^p(\mu)$ is a Banach space for every $1 \leq p \leq \infty$. For $p = 2$ it is a Hilbert space with inner product $\langle f \rangle g = \int f \bar{g} d\mu$.

Quick-Reference: Key Inequalities

Name	Statement	Conditions
Markov	$\mu(f \geq \lambda) \leq \lambda^{-1} \ f\ _1$	$f \in L^1, \lambda > 0$
Chebyshev	$\mu(f \geq \lambda) \leq \lambda^{-2} \ f\ _2^2$	$f \in L^2, \lambda > 0$
Hölder	$\ fg\ _1 \leq \ f\ _p \ g\ _q$	$\frac{1}{p} + \frac{1}{q} = 1$
Minkowski	$\ f + g\ _p \leq \ f\ _p + \ g\ _p$	$1 \leq p \leq \infty$
Jensen	$\phi(\int f d\mu) \leq \int \phi \circ f d\mu$	ϕ convex, μ a prob. measure